

A focal loss and sequential analytics approach for liver disease classification and detection

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ARTICLE INFO

Handling Editor: Madijd Tavana

Keywords:

Attention mechanism analytics
Sequential pattern recognition
Imbalanced data handling
Predictive model evaluation
Clinical decision analytics
Medical data interpretation

ABSTRACT

Liver disease poses a significant global health challenge requiring accurate and timely diagnosis. This research develops a novel deep learning model, named AFLID-Liver, to improve the classification of liver diseases from medical data. The AFLID-Liver model integrates three key techniques: an Attention Mechanism to focus on the most relevant data features, Long Short-Term Memory (LSTM) networks to process potential sequential information, and Focal Loss to effectively handle imbalances between different disease classes in the dataset. This combination enhances the model's ability to learn complex patterns and make robust predictions. We evaluated AFLID-Liver using a dataset of various patient records, including biomarkers and demographics. Our proposed model achieved superior performance, with 99.9 % accuracy, 99.9 % precision, and a 99.9 % F-score, significantly outperforming a baseline Gated Recurrent Unit (GRU) model (99.7 % accuracy, 97.9 % F-score) and existing state-of-the-art approaches. These results demonstrate AFLID-Liver's potential for highly accurate liver disease detection. To validate the generalizability of the proposed model, we performed cross validation using an external dataset which also yielded a good performance depicting the potential of the proposed model. The novelty lies in the synergistic integration of these techniques, offering a robust approach for clinical decision support and improved patient outcomes. Future research will aim to enhance the computational efficiency, paving the way for its adoption in real-time clinical applications.

1. Introduction

Liver diseases represent a significant global health burden, encompassing a diverse array of conditions that affect the liver, a vital organ essential for metabolic, detoxification, and digestive processes [1–4]. These conditions range from viral infections such as hepatitis to lifestyle-related diseases like alcoholic liver disease (ALD) and non-alcoholic fatty liver disease (NAFLD) [5–8]. Complications including cirrhosis and hepatocellular carcinoma (HCC) account for a substantial proportion of liver disease-related morbidity and mortality [9,10]. According to the World Health Organization (WHO), over 300

million people globally live with chronic hepatitis B or C, with more than 1.3 million annual deaths attributed to these infections [11–13]. The increasing prevalence of NAFLD and alcohol-related liver disease further highlights the urgency of advancing diagnostic and management strategies for earlier prediction and diagnosis [14–16].

Accurate and timely diagnosis of liver diseases is crucial for improving patient health, yet it remains a challenging task due to overlapping symptoms, variability in disease progression, and reliance on invasive procedures like biopsies [17–19]. As depicted in Fig. 1, there exist invasive and noninvasive techniques for liver disease diagnosis. Researchers have explored various approaches to improve liver disease

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<https://doi.org/10.1016/j.health.2025.100424>

Received 13 February 2025; Received in revised form 1 May 2025; Accepted 2 October 2025

Available online 4 October 2025

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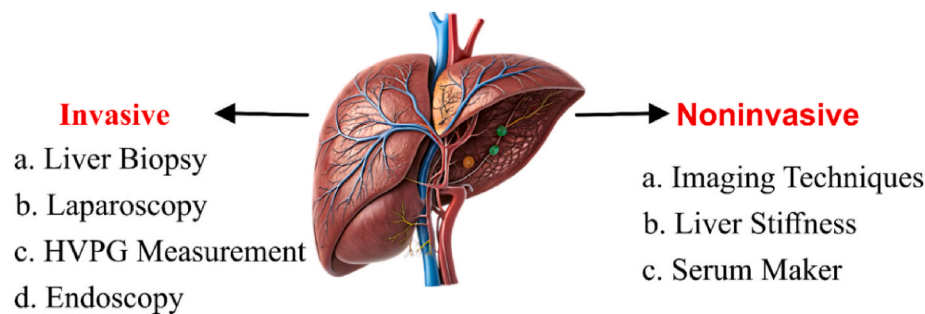


Fig. 1. Invasive and noninvasive techniques for liver disease diagnosis [41].

prediction, employing both traditional and technology-driven methodologies [20–24]. Earlier methods primarily relied on clinical examinations, such as liver biopsies, which, while accurate, are invasive and time-consuming. Recently, advancements in computational techniques such as machine learning (ML) and deep learning (DL) have revolutionized the field, offering innovative solutions to overcome diagnostic challenges [25,26].

ML and DL techniques have emerged as transformative tools in liver disease classification, leveraging large datasets to identify patterns and provide predictions with greater accuracy and efficiency [19,27,28]. These approaches have been applied to diverse datasets, including biochemical markers, imaging data, and patient demographics, to aid in the detection, staging, and management of liver diseases [29–31].

Among these innovations are supervised ML models designed to predict liver disease stages or classify conditions using diverse data sources, including biochemical, genetic, and imaging datasets [32]. Convolutional Neural Networks (CNNs) is a technique that has been used in the classification, segmentation, and registration of various medical images, including endoscopic images, X-rays, magnetic resonance imaging (MRI), computed tomography (CT), ultrasounds, skin imagery, and histopathology slides. In liver diagnostics, CNNs have proven particularly effective in analyzing ultrasound and CT scans, enabling the detection of structural abnormalities and enhancing the accuracy of liver disease assessments [33,34]. Despite the fact that CNNs have been widely used for liver disease classification, yet they come with notable limitations such as overfitting, data imbalance, computational requirements and feature learning limitations [35,36]. On the other hand, recurrent neural networks (RNNs) have been utilized to process time-series data, such as variations in liver enzyme levels, enabling dynamic disease monitoring [37].

Moreover, researchers have focused on feature engineering, emphasizing the selection and extraction of the most significant biomarkers from large, intricate datasets [38–40]. This approach enhances model accuracy and interpretability, facilitating the development of more robust and clinically relevant diagnostic tools. These advancements collectively demonstrate the potential of leveraging ML and DL to transform liver disease diagnostics, making them less invasive, cost effective, more accurate, and accessible.

However, several research gaps hinder the broader application of ML and DL in liver disease diagnostics. Existing studies report up to 99 % accuracy, while they often overlook the performance on minority classes, critical in liver disease classification. Our study prioritizes balanced prediction, improving metrics like F1-score & ROC which previous works have not adequately addressed due to class imbalance issues. These include the need for advanced data augmentation techniques to address dataset imbalance, robust feature selection and extraction methods to enhance model interpretability, and effective handling of missing data and outliers to ensure reliability. Additionally, optimization of ML algorithms, integration of multi-modal data, and the adoption of comprehensive performance metrics are critical for improving the utility and clinical relevance of these models. Addressing these gaps through innovative approaches will not only advance the field of liver

disease diagnostics but also pave the way for more personalized and efficient patient care.

This study aims to propose a novel model called AFLID-Liver for liver disease classification by addressing critical limitations identified in previous research. It will leverage attention mechanism combined with Long Short-Term Memory (LSTM) networks to tackle challenges associated with limited datasets. By integrating Focal Loss, the approach will effectively mitigate class imbalance, enabling more accurate and equitable predictions across all categories. Additionally, the research will conduct comparative analyses with state-of-the-art GRU technique to identify their relative strengths and limitations. To enhance generalizability, the model will undergo rigorous testing, ensuring their applicability in broader clinical settings. By bridging the gap between innovative AI methodologies and real-world challenges, this study aims to improve diagnostic accuracy, advance patient care, and foster the adoption of artificial intelligence in healthcare.

Research contributions:

- We proposed an attention-driven LSTM model enhanced with focal loss (AFLID-Liver) to improve the classification of liver disease, effectively focusing on critical features and mitigating the challenges of imbalanced datasets.
- We designed a hybrid neural network combining Attention Mechanism (AM) and Long Short-Term Memory (LSTM) to extract and train the key features of liver disease data. The attention mechanism enables the model to focus on specific critical areas of the input sequence, enhancing feature representation. This hybrid approach improved the accuracy of liver disease classification while also reducing training and prediction time.
- We designed a focal loss function tailored to address class imbalance in liver disease data, reducing bias toward majority classes and significantly improving classification accuracy for minority disease instances.

Motivation for the study

Key Insight	Description
Problem/Issue	Early and accurate detection of liver diseases is critical for timely medical intervention, yet existing models often struggle with class imbalance and sequential feature dependencies, leading to suboptimal performance.
Existing Solutions	Existing methods, including GRU-based models, have been employed for liver disease classification, achieving reasonable accuracy. However, they face limitations in addressing class imbalance and leveraging sequential data effectively
Novel Contribution	This study introduced AFLID-Liver, a novel model combining Attention Mechanism (AM), Long Short-Term Memory (LSTM), and Focal Loss (FL). The model achieved 99.9 % accuracy and F-score, surpassing GRU-based models and other state-of-the-art approaches. By addressing class imbalance and

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Key Insight	Description
Beneficiaries of the Proposed Solution	capturing sequential dependencies, AFLID-Liver provides a robust and precise solution for liver disease classification.
	Clinicians and medical researchers focused on liver disease diagnosis would benefit from the enhanced accuracy and robustness of AFLID-Liver. Additionally, AI researchers in medical imaging and classification can build upon the methodologies introduced in this paper.

2. Related works

Machine learning (ML) has revolutionized healthcare, offering solutions to critical challenges such as optimizing cancer treatment, early detection of liver diseases, and accurate diagnosis of conditions like dengue fever. Liver disease classification, in particular, has emerged as a significant area of research, with advanced ML algorithms being employed to enhance diagnostic accuracy and improve patient health. Among notable research in this area is a study conducted by Ref. [42], who introduced an optimized ensemble machine learning framework (ONBLR) for liver cirrhosis classification. The study leveraged Grasshopper Optimization Algorithm (IGOA) for feature selection. Classification was performed using a hybrid ensemble model combining Naïve Bayes and Logistic Regression, optimized with the Harris Hawks Optimization Algorithm. The method achieved high accuracy (99.21 % and 99.11 %) and sensitivity on two datasets, outperforming baseline models like SVM and Random Forest. Strengths include improved accuracy, effective feature selection, and computational efficiency. However, limitations involve reliance on small datasets, potential overfitting, and challenges with continuous features in Naïve Bayes. Future research could expand the approach to broader liver disease categories for better generalizability.

While [42]’s work focused on cirrhosis classification, other researchers have explored different machine learning strategies for liver disease. For instance, the researchers in Ref. [43] evaluated the performance of seven boosting algorithms: Gradient Boosting (GB), AdaBoost (AB), LogitBoost (LB), Stochastic Gradient Boosting (SGB), XGBoost (XGB), LightGBM (LGBM), and CatBoost (CB) for predicting chronic liver disease (CLD) using two datasets: Liver Disease Patient Dataset (LDPD) and Indian Liver Disease Patient Dataset (ILPD). The study involved Exploratory Data Analysis (EDA), normalization, SMOTE for class balancing, and hyperparameter optimization. GB achieved the highest accuracy (98.80 % for LDPD and 98.29 % for ILPD) but was computationally intensive, while LGBM was the fastest but slightly less accurate. The research highlights the potential of boosting algorithms for improving CLD prediction but notes limitations like computational overhead and false positives. In Ref. [44], the authors introduced a hybrid deep learning model combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks for liver disease diagnosis and subtyping. The model, trained on the Kaggle liver patient dataset, used CNN for spatial feature extraction and LSTM for temporal pattern recognition. It achieved 98.73 % accuracy, 99 % precision, 98 % recall, 98 % F1 score, and 99 % AUC-ROC, outperforming standalone CNN, RNN, and LSTM models. However, the model faces limitations such as misclassification and high computational complexity.

Beyond boosting and hybrid models, ensemble learning has also proven effective in tackling liver disease classification. The researchers in Ref. [45] investigated an ensemble learning framework with advanced preprocessing techniques for liver disease classification using the Indian Liver Patient Dataset (ILPD). Six ensemble algorithms: Random Forest, Gradient Boosting, Bagging, XGBoost, Extra Trees, and

Stacking were evaluated, with hyperparameter optimization to enhance performance. Results showed the Extra Trees classifier achieved the highest testing accuracy of 91.82 %, followed by Random Forest at 86.06 %. The stacking model excelled in 10-fold cross-validation, achieving 93.15 % accuracy. Despite the robust preprocessing pipeline and strong classifier performance, limitations include low performance on minority classes, potentially limiting generalizability, and the computational complexity of ensemble methods.

Further demonstrating the breadth of ML applications in this field [46], explored a machine learning framework for diagnosing and classifying liver diseases using five classification algorithms: Regularized Regression (RR), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). The dataset of 525 patients, comprising 18 biochemical and hematological risk factors, underwent preprocessing steps like missing value imputation, normalization, and feature selection. Random Forest achieved the best performance with 76.2 % accuracy, 84.3 % recall, a 77.5 % F1-score, and an AUC of 0.999. The study emphasizes cost-effective diagnostic tools and highlights critical risk factors, making it particularly valuable for low-resource regions. However, limitations include moderate accuracy, suggesting scope for improvement.

While many studies focus on traditional clinical data for diagnosis, others delve into the molecular underpinnings of liver conditions [47]. investigated the role of cuproptosis-related genes (CRGs) in non-alcoholic fatty liver disease (NAFLD), identifying molecular clusters linked to immune cell infiltration and diagnostic biomarkers. Data from the GEO database were preprocessed, and nine differentially expressed CRGs were used to cluster NAFLD samples into two groups. Weighted Gene Co-expression Network Analysis (WGCNA) identified cluster-associated genes, and machine learning models like SVM and Random Forest predicted molecular clusters, with SVM achieving an AUC of 0.950. Three biomarkers: SLC16A1, ENO3, and LEPR were validated, showing significant expression differences between NAFLD and control groups. Validation using a high-fat diet mouse model and in vitro experiments confirmed increased lipid accumulation and altered gene expression. Strengths include integrating clustering, machine learning, and experimental validation. Nevertheless, the method tends to perform low on huge data, which may restrict generalizability.

Returning to algorithmic performance evaluations, a comparative analysis of seven boosting algorithms was conducted by the authors [48] for chronic liver disease prediction, where Gradient Boosting (GB) proved to be the top performer across both datasets used, achieving accuracy rates of 98.8 % on LDPD and 98.29 % on ILPD. Additionally, GB surpassed other boosting algorithms in most performance metrics, with the exception of runtime. In a similar study, the researcher [49] proposed a machine learning approach for liver cancer diagnosis based on gene expression data, integrating feature selection methods with a stacking ensemble learning model. The approach tackles the challenges posed by high dimensionality and intricate patterns in genomic data, aiming to enhance diagnostic precision and interpretability [50]. presents a system for predicting liver disease (LD) by integrating machine learning (ML) algorithms such as Decision Tree, Random Forest, Extra Trees Classifier (ETC), LightGBM, and AdaBoost, along with the Tree-Structured Parzen Estimator (TPE) for hyperparameter optimization.

Expanding the scope to include medical imaging data and broader classifier comparisons, the study in Ref. [51] presented a modified Unet-60 deep learning model for detecting and classifying liver tumors (metastasis carcinoma and cholangiocarcinoma) using 864 annotated CT images from the 3Dircadb database. Preprocessing included resizing and segmentation, with feature selection using a binary differential evolution algorithm. The model achieved high performance metrics, including 98.61 % accuracy and 100 % specificity. Meanwhile, the study

by Ref. [52] evaluated 10 machine learning classifiers for liver disease diagnosis using datasets from UCI and GitHub. Preprocessing involved missing value replacement and attribute conversion. Random Forest (RF) achieved the highest accuracy (72.17 %) on the UCI dataset, while SVM performed best (71.36 %) on the GitHub dataset. Furthermore, Researchers [53] examined the effectiveness of ensemble learning methods (boosting, bagging, voting) for predicting liver disease using a clinical dataset of 30,691 samples. Gradient Boosting (GB) achieved the best accuracy (98.80 %) but had slower runtimes compared to methods like LightGBM. Limitations include potential biases from SMOTE for data balancing and reliance on mean-median imputation.

Exploring alternative diagnostic modalities, another study by the authors [54] investigated the feasibility of using medical infrared thermal imaging (MITI) for the classification of non-alcoholic fatty liver disease (NAFLD) using a dataset of 167 thermal images from 32 patients. Preprocessing involved foreground extraction, histogram equalization, and manual selection of liver-specific regions of interest (ROI). Features were extracted using texture analysis and pre-trained CNNs like Inception-ResNet-v2 and DenseNet-201, with PCA for feature selection. Classification achieved 94 % accuracy with CNN-based features and filtered classification methods, with AUC values ranging from 0.69 to 0.92. While MITI offers a non-invasive, cost-effective diagnostic alternative, limitations include limited performance in imbalanced scenarios and reliance on manual ROI selection.

To provide a consolidated overview of these diverse research efforts, Table 1 provides a comprehensive summary of the key aspects of these studies, highlighting the methods, class-balancing techniques, and their limitations.

3. Methodology

3.1. Attention-enhanced LSTM with focal loss

This section outlines the proposed approach for liver disease classification, as illustrated in Algorithm 1. The method utilizes the sequence modelling capabilities of Long Short-Term Memory (LSTM) networks, combined with an attention mechanism, to prioritize critical features and temporal patterns within the medical data. The attention mechanism ensures that the model focuses on the most relevant aspects of the data, improving its ability to detect subtle variations in liver disease patterns. Additionally, Focal Loss was incorporated to address class imbalance, particularly enhancing the detection of rare disease

Table 1
Summary of studies overview.

Ref.	Year	Methods	Class Balancing Technique	Limitation
[42]	2024	Naïve Bayes + LR	NIL	Scalability issues, risk of overfitting, and difficulty handling continuous features in Naïve Bayes.
[43]	2024	Gradient Boosting	SMOTE	High computational overhead and likelihood of lack of generalizability.
[44]	2024	CNN + LSTM	NIL	Dependency issue and high computational complexity due to the hybrid architecture.
[45]	2023	Extra Tree Classifier	SMOTE	Likelihood of synthetic data generated not reflecting actual data leading to reduce performance.
[46]	2022	Random Forest	NIL	Low accuracy.
[53]	2024	Gradient Boosting	SMOTE	Reliance on SMOTE for data balancing and mean-median imputation methods may introduce biases. Slower runtime in Gradient boosting.

instances, which are often underrepresented in the dataset. The following subsections provide a detailed description of the model architecture, demonstrating how these components are integrated to achieve accurate and efficient liver disease classification. The architecture of the proposed model is presented in Fig. 2.

3.2. Data preprocessing

In the pre-processing stage, we screened the data by removing null records and duplicate entries and applied the following technique to scale all attributes for consistent modeling.

a. Min-Max Scaling

The Min-Max normalization technique [55] transforms a feature's values to fall within the range [0, 1]. This method ensures that the data is scaled consistently, enhancing its suitability for the classification task. The mathematical formula for Min-Max scaling is presented in Equation (1).

$$J_{new} = \frac{J_{new} - \min(j)}{\max(j) - \min(j)} \quad (1)$$

Where J_{new} represents the normalized value of a specific feature, while $\min(j)$ and $\max(j)$ denote the feature's lowest and highest values, accordingly.

b. Standard Scaling (Z-score Normalization)

Standard Scaling transforms the data such that the mean is 0 and the standard deviation is 1 [55]. This is done using the following formula presented in Equation (2).

$$J_{new} = \frac{J - \mu}{\sigma} \quad (2)$$

Where J_{new} is the standardized value, J is the original value, while μ is the mean of the feature, and σ is the standard deviation of the feature.

c. Robust Scaling

Robust Scaling is designed to be robust against outliers. It scales the data based on the median and the interquartile range (IQR) [55]. The formula is depicted in equation (3)

$$J_{new} = \frac{j - \text{median}(X)}{\text{IQR}(X)} \quad (3)$$

Where $\text{median}(X)$ is the median of the feature, $\text{IQR}(X) = Q3 - Q1$ is the interquartile range (the difference between the third quartile $Q3$ and the first quartile $Q1$).

3.3. Long Short-Term Memory (LSTM)

LSTM is a specialized architecture of recurrent neural networks (RNNs) [56] designed to address long-term dependencies by incorporating mechanisms such as the forget gate, input gate, and output gate. These gates regulate the flow of information, allowing the model to remember, update, or discard data as needed. In the context of liver disease classification, the LSTM architecture captures temporal dependencies in medical data, such as patient history or time-series biomarkers, to model the progression of diseases effectively. The output gate establishes the final output of the current node, the forget gate allows the neural network to exclude extraneous information, and the input gate feeds the neural network fresh data.

The following equation describes how the LSTM model propagates forward. The LSTM model's two hidden states are represented by h_t and C_t . Sigmoid and hyperbolic tangent functions are denoted by the

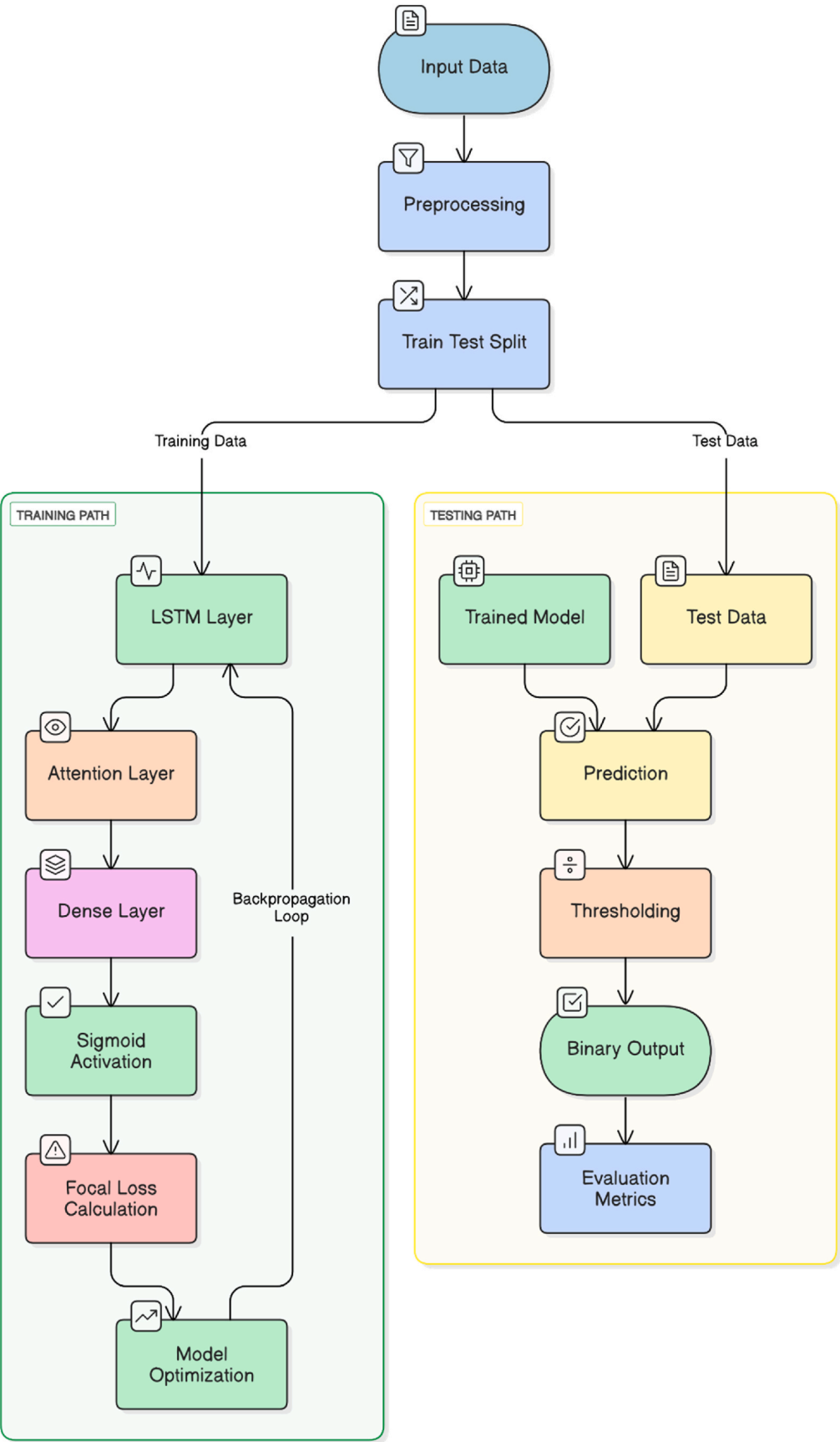


Fig. 2. Architecture of the proposed model.

symbols σ and $\tanh$. The input, forget, and output gate are represented by the variables i, f , and o . Weight matrices connected to various peephole connections are represented by the term W . The forget gate decide which information from the previous cell state should be discarded, the forget gate's output is updated according to equation (4):

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

Where f_t is the forget gate's output, h_{t-1} represents the hidden state from the previous step, x_t is the current input, and W_f, b_f are the weight matrix and bias.

Next is to update the cell state with new information as depicted in equations (5) and (6):

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (6)$$

Here i_t determines the degree to which new information will influence the cell state, and $\tilde{C}_t$ is the candidate cell state. Subsequent, the forget and input gates are combined to update the memory cell state as illustrated in equation (7):

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (7)$$

Then, the output at the current time step is determined, and updated accordingly, as found in equations (8) and (9):

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (8)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (9)$$

By capturing temporal dependencies, LSTM effectively models sequences of medical data and identifies correlations between past and current patient health metrics, which is particularly crucial for detecting patterns indicative of liver disease. LSTM has the capability not only to analyze the current medical data but also to retain information about past health indicators. In the context of liver disease, certain symptoms

or diagnostic markers may develop gradually over time. Consequently, current medical observations might appear normal or may not exhibit significant connections to earlier health data, making LSTM essential for uncovering subtle and progressive disease patterns.

3.4. Attention mechanism

The attention mechanism enhances the LSTM's performance by allowing the model to focus on the most relevant features or time steps in the sequence [57]. This is particularly beneficial for liver disease classification, where certain features (e.g., elevated biomarkers or specific patterns in imaging data) may hold more diagnostic significance. The attention weight α_t for a given time step t is computed as seen in equation (10):

$$\alpha_t = \frac{\exp(u_t^T u_w)}{\sum_t \exp(u_t^T u_w)} \quad (10)$$

Where u_t is the hidden state representation at time t , and u_w is the learnable weight vector [58]. The hidden state representation u_t is calculated using equation (11):

$$u_t = \tanh(W_w \cdot h_t + b_w) \quad (11)$$

where, W_w and b_w are learnable parameters representing the weight matrix and bias. Once the values for the attention probability distribution at each moment have been determined, we compute the weighted feature vector v , which contains information about the patients record, using equation (12):

$$v = \sum_t \alpha_t \cdot h_t \quad (12)$$

Lastly, to obtain the predicted label y , we applied the softmax function, depicted in equation (13):

$$y = \text{softmax}(W_v \cdot v + b_v) \quad (13)$$

Where, W_v and b_v are the weight matrix and bias for classification.

The attention-enhanced LSTM model combines the context vector with the LSTM's outputs to emphasize critical features, improving classification accuracy.

3.5. Class-wise focal loss (FL)

Class imbalance is a significant challenge in liver disease classification, as the number of patients with certain rare conditions may be much smaller than those with common diseases. To address this, the focal loss function was employed, which modifies the standard cross-entropy loss to focus more on minority sample instances [59,60]. The loss for each class is defined in equation (14) as:

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (14)$$

Where α_t is the class weight, γ is the focusing parameter, and p_t is the predicted probability for the true class. For a batch size N , the total loss was computed in equation (15) as:

$$L_{total} = \frac{1}{N} \sum_{i=1}^N FL(p_t^{(i)}) \quad (15)$$

We incorporated mini-batch stochastic gradient descent and the Adam optimizer to update the model's weights iteratively during training. This approach ensures that minority class instances receive greater attention during training, leading to improved sensitivity for rare liver diseases.

Table 2
Distribution of liver patient dataset (LDP).

S/ N	Features	Data Types	Description
1.	Age	Integer	Represents the age of the patient in years
2.	Gender	String	Indicates the biological sex of the patient
3.	Total Bilirubin (TB)	Float	A measure of bilirubin in the blood, indicating the breakdown of red blood cells.
4.	Direct Bilirubin (DB)	Float	A conjugated form of bilirubin, suggesting liver processing
5.	Alkaline Phosphatase (Alkphos)	Integer	An enzyme present in the liver and other tissues
6.	Alamine Aminotransferase (SGPT)	Integer	An enzyme primarily found in the liver, often elevated in liver damage.
7.	Aspartate Aminotransferase (SGOT)	Integer	An enzyme found in the liver and other organs, also indicative of potential liver damage
8.	Total Proteins (TP)	Float	A measure of the total protein content in the blood
9.	Albumin (ALB)	Float	A major protein produced by the liver
10.	Albumin and Globulin Ratio (A/G Ratio)	Float	Reflects the balance between albumin and other proteins in the blood
11.	Target	Boolean	Classifies individuals as either "Liver Patient" or "Non-Liver Patient"

Algorithm 1: Attention-Enhanced LSTM with Focal Loss (Liver Disease Classification)**Input:**

Training Data: X_{train}, Y_{train} ,
 Hyperparameters: $L_{hidden\ layer}, W_{attention}, \gamma_{FL\ focusing_param}, \alpha_{class_weigh},$
 $N_{batch_size}, l_{earning_rate}, e_{epochs}, A_{optimizer}$

Output: Trained Attention-Enhanced LSTM model θ

Initialization:

1. Define the Model Architecture
 - LSTM for sequence modeling
 - Attention Mechanism to assign importance to time steps
 - Fully connected classification layer
2. Define Focal Loss Function:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$
3. Initialize Weight matrix $W_{attention}$ for attention mechanism
 for $k = 1, 2, \dots, n, i = 1, 2, \dots, m$

Training Process:

Repeat until convergence:

For each epoch $t = 1$ to e_{epochs} :

a. Mini Batch Sampling:

Divide X_{train}, Y_{train} , into mini-batches of size N

b. Forward Pass:

For each mini-batch (X, Y) :

Compute LSTM outputs O_{lstm} :

$$O_{lstm} = \text{LSTM}(X)$$

Compute attention weights for each sequence:

$$W_{attention} = \text{softmax}(W \cdot O_{lstm})$$

Calculate the attention-weighted representation:

$$O_{attention} = W_{attention} \cdot O_{lstm}$$

Generate predictions using the fully connected layer:

$$\hat{Y} = \text{Softmax}(\text{FC}(W \cdot O_{lstm}))$$

c. Compute Focal Loss:

For each class c in Y :

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

Compute the total batch loss:

$$L_{batch} = \frac{1}{N} \sum_{i=1}^N FL(p_t^{(i)})$$

d. Backpropagation and Weight Updates:

Backpropagate the loss L_{batch} to update:

LSTM weights

Attention weight matrix $W_{attention}$

Fully connected layer parameters

Use $A_{optimizer}$ to update all trainable Parameters

Increment epoch: $t = t + 1$.

Evaluation:

Evaluate the trained model θ using test data (X_{test}, Y_{test})

Compute performance metrics:

Accuracy, Precision, Recall, F1-Score, and False Positive Rate (FAR)

Output:

Return the trained Attention-Enhanced LSTM model θ

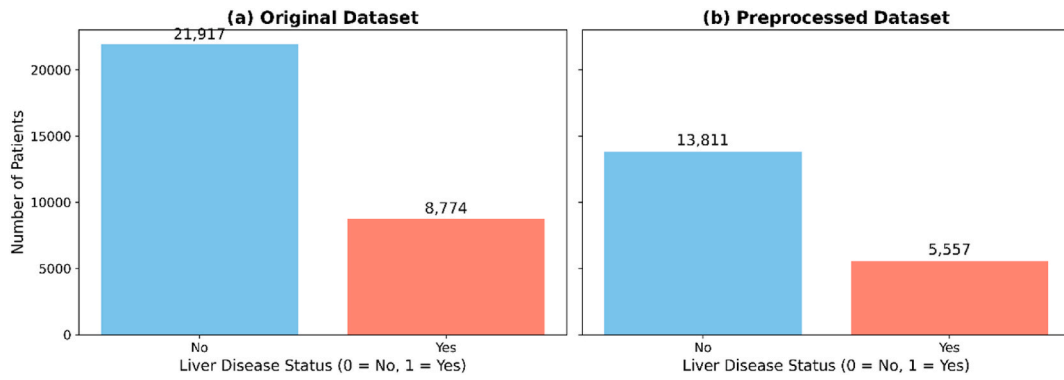


Fig. 3. Distribution of Dataset according to target class.

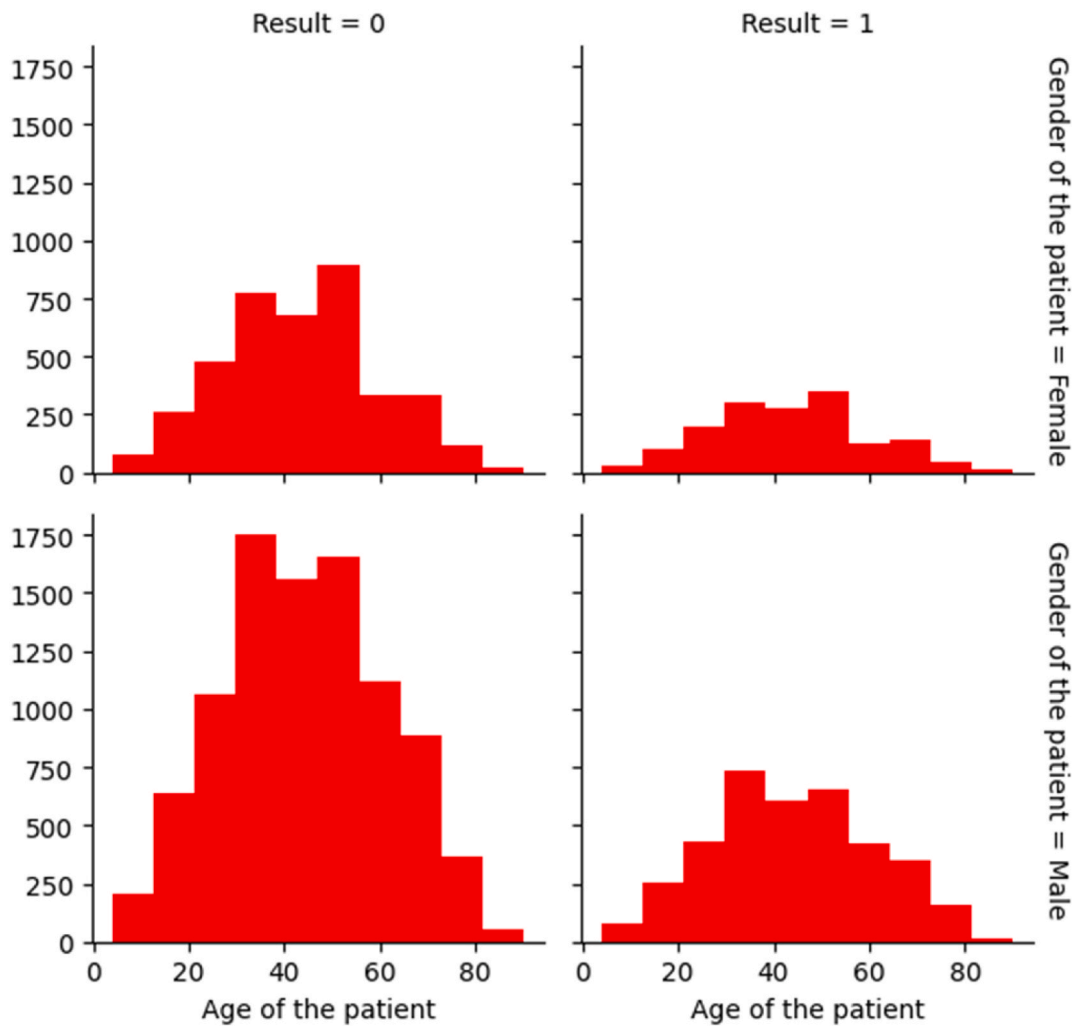


Fig. 4. Disease by gender and age.

4. Experimental evaluations

This section begins by presenting an overview of the experimental setup, including the evaluation metrics and dataset used. Subsequently, an in-depth analysis of the proposed model's performance was provided. Finally, the ablation study was presented to demonstrate the effectiveness of our proposed approach.

4.1. Dataset description

The study utilized the Liver Patient Dataset, sourced from Kaggle (<https://www.kaggle.com/datasets/abhi8923shriv/liver-disease-patient-dataset>). This dataset comprises 11 attributes and 30,691 patient records, encompassing demographic factors (age, gender) and biochemical markers related to liver function. The biochemical markers include Total Bilirubin (TB), Direct Bilirubin (DB), Alkaline Phosphatase (Alkphos), Alanine Aminotransferase (SGPT), Aspartate

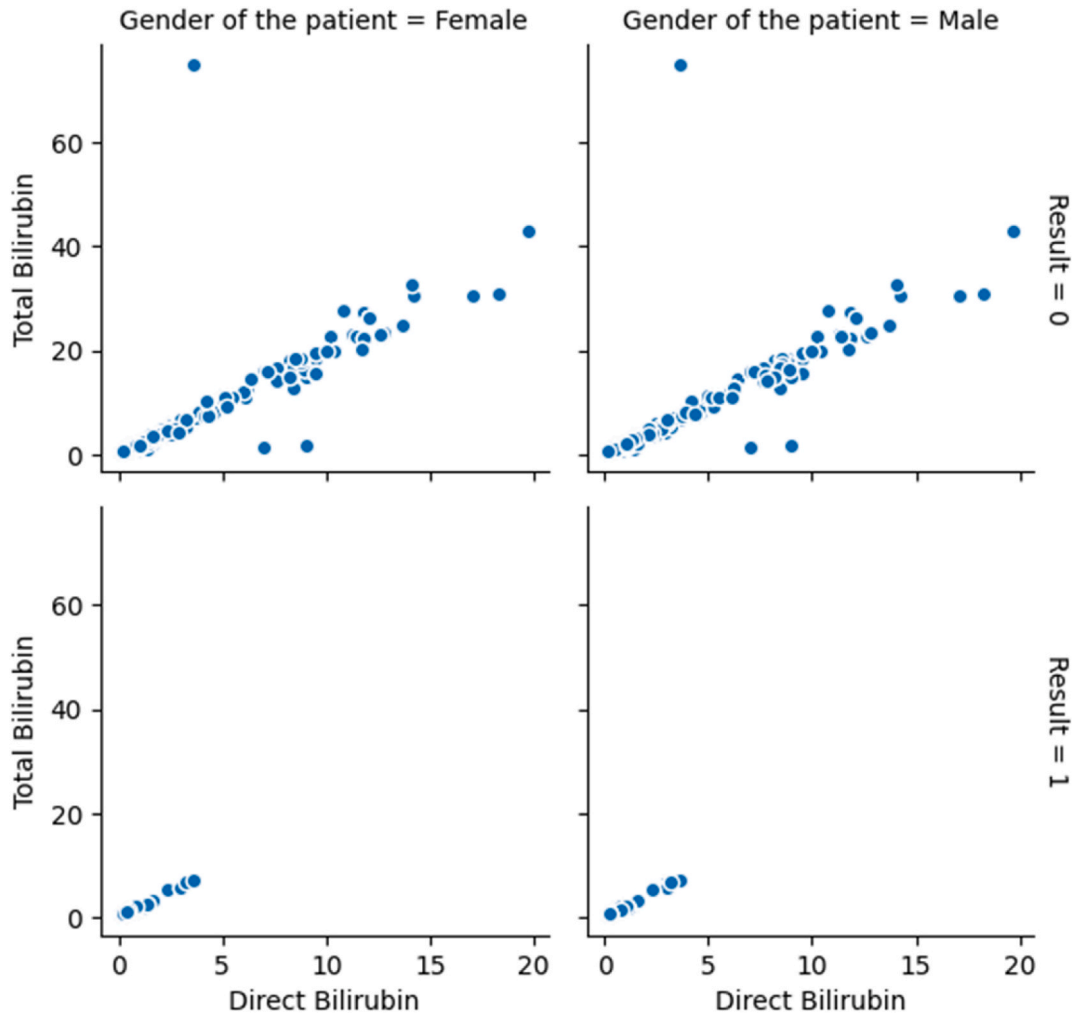


Fig. 5. Relationship between direct and total bilirubin, stratified by gender and liver disease status.

Aminotransferase (SGOT), Total Proteins (TP), Albumin (ALB), and Albumin-Globulin Ratio (A/G Ratio). The target variable, annotated by medical experts, classifies individuals into two categories: "Liver Patient" or "Non-Liver Patient." For the purpose of this study, the target variable was re-encoded as 0 for "Liver Patient" and 1 for "Non-Liver Patient." The dataset has 11,323 number of duplicate rows, after removing the duplicate records, we have 19,368 records which was processed for experimentation. A detailed description of the dataset is provided in Table 2. Distribution of the dataset preprocessing and postprocessing is depicted in Fig. 3.

Fig. 3(a) original dataset and Fig. 3(b) shows dataset distribution after preprocessing stage.

Fig. 4 displays liver disease prevalence by gender and age using a 2×2 facet grid of histograms. The grid separates patients by liver disease status (Result = 0: no disease; Result = 1: disease) and gender (Female: top row; Male: bottom row), showing patient counts versus age. Observations indicate that the age distribution for both genders, with and without liver disease, peaks between 40 and 60 years, particularly around 40–50 years in non-diseased individuals. Liver disease patients similarly concentrate in the 40–60 age range, suggesting this period as a high-risk window. A notable gender disparity is observed, with a substantially higher number of male patients diagnosed with liver disease compared to females, aligning with established epidemiological trends.

Fig. 5 presents a 2×2 facet grid of scatter plots showing the relationship between Direct and Total Bilirubin, stratified by patient gender and liver disease status. In individuals without liver disease (Result = 0),

both genders exhibit a strong positive correlation between Direct and Total Bilirubin at low concentrations, consistent with normal bilirubin metabolism. Conversely, patients with liver disease (Result = 1) display generally higher and more variable bilirubin levels, suggesting impaired hepatic processing or biliary excretion. The plots highlight the expected elevation and altered relationship between Direct and Total Bilirubin as a biochemical hallmark of liver dysfunction, visualized across male and female patient subgroups.

Fig. 6 is a heatmap of the Pearson correlation matrix for various patient features, including biological markers and demographics. It visually represents the linear relationship between each pair of variables, with color intensity and value indicating the strength and direction (positive/red, negative/blue) of the correlation. Scientifically, this matrix reveals expected strong positive correlations between related biomarkers like Total and Direct Bilirubin (0.90), and Total Proteins and Albumin (0.78), reflecting their biochemical interdependencies. Conversely, it shows weaker correlations between demographic factors like age and most biological measures. This visualization serves to quickly identify patterns of association within the dataset, aiding in understanding feature relationships and informing subsequent analyses.

4.2. Metrics used for performance evaluation

To evaluate the performance of the proposed model in classifying liver disease within the imbalanced patient data, we utilized six evaluation metrics: accuracy (ACC), precision (PRC), recall (RCL), F-score,

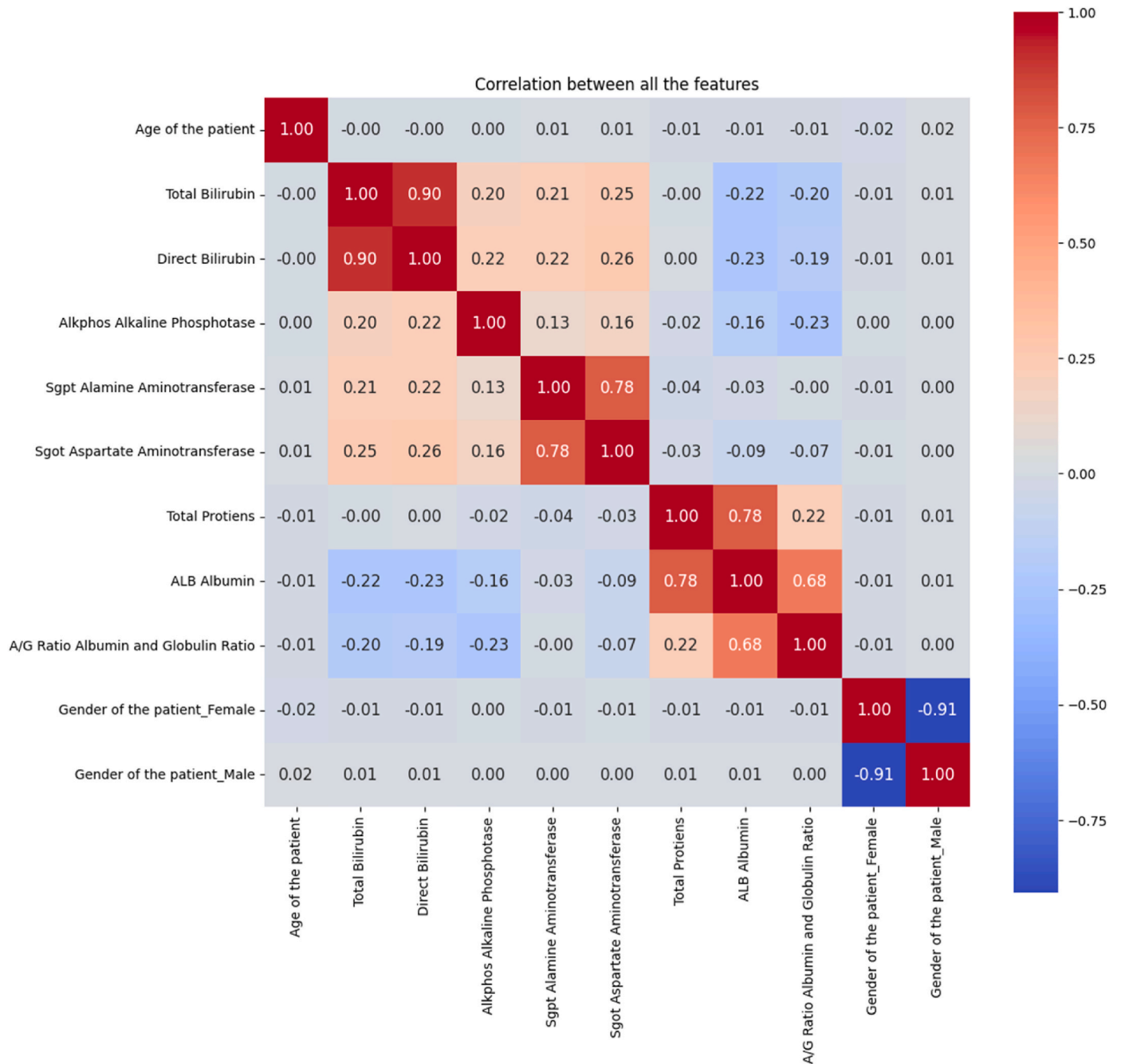


Fig. 6. heatmap of the Pearson correlation matrix.

false positive rate (FPR), false negative rate (FNR), area under the receiver operating characteristic curve (AUC-ROC), matthews correlation coefficient (MCC), area under the precision-recall curve (AUPRC). In this context, True Positive (TP) represents the number of cases where the model correctly identifies individuals with liver disease. Conversely, False Negative (FN) refers to the cases where individuals with liver disease are incorrectly classified as non-liver patients by the model. True Negative (TN) denotes the cases where the model accurately identifies individuals without liver disease. Meanwhile, False Positive (FP) describes instances where the model incorrectly classifies individuals without liver disease as liver patients.

Accuracy is the proportion of correctly classified instances (both liver and non-liver patients) out of the total instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (16)$$

Precision is the model's ability to correctly identify liver disease cases out of all instances predicted as liver disease.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (17)$$

Recall indicates the model's effectiveness in identifying actual liver disease cases by measuring the proportion of true positives out of all actual liver disease cases

$$\text{Recall} = \frac{TP}{TP + FN} \quad (18)$$

F-Score provides a balanced measure of the model's performance by

Table 3

Design principles of proposed model.

	Parameters	Value
LSTM-AM	Epochs	600
	Input Layers (Shape)	7
	LSTM Layer (Units)	128
	Batch Size	32
	Dropout	[0.1, 0.2, 0.3, 0.4, 0.5]
	Learning Rate	[0.001, 0.0008, 0.0005, 0.0003]
	Activation Function	ReLU (Dense Layer) Sigmoid (Output Layer)
	Attention Weight	Softmax
FL	Optimizer	Adam
	Alpha Value	0.25
	Gama Value	2.0

combining precision and recall

$$F - \text{Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (19)$$

FPR reflects the proportion of non-liver disease cases that were incorrectly classified as liver disease

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (20)$$

FNR represents the proportion of liver disease cases that the model failed to identify correctly

$$\text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}} \quad (21)$$

AUC-ROC measures the ability of a binary classifier to distinguish between two classes across all possible classification thresholds. The ROC curve plots the (TPR) against the FPR at various threshold settings.

AUC Interpretation is as follows: perfect classifier (AUC = 1.0), random guessing (AUC = 0.5), worse than random (AUC < 0.5)

MCC is a single-value metric that measures the quality of binary (or multiclass) classifications, considering all four confusion matrix categories: TP, TN, FP, and FN. It is essentially the correlation coefficient between observed and predicted binary classifications, MCC is presented in equation (22)

$$\text{MCC} = \frac{(TP * TN - FP * FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (22)$$

Where MCC = 1 is perfect prediction, MCC = 0 is random prediction,

MCC = -1 is total disagreement.

AUPRC quantifies the area under the Precision-Recall (PR) curve, which plots Precision versus Recall at various thresholds. AUPRC is Interpret as follows Higher AUPRC indicates a better classifier.

4.3. Experimental settings and model configuration

The simulations for the proposed model were conducted on Google Colab, utilizing GPUs with 12 GB RAM for efficient computation. Python 3.9 served as the primary programming language for the experimental environment. The model was implemented using TensorFlow, Keras, and Scikit-Learn, while visualization and data analysis were supported by Seaborn, Matplotlib, NumPy, and Pandas. Table 3 outlines the parameter settings for the proposed model. Extensive experiments were performed, varying learning rates, epochs, gamma values, and alpha values, to optimize performance. The results demonstrated that the model's effectiveness is highly sensitive to the selection of these parameters. To address the class imbalance in the dataset, the focal loss (FL) function was integrated into the LSTM model. Gamma and alpha values were carefully fine-tuned for different classes, showcasing the model's flexibility and adaptability in handling imbalanced data. This comprehensive parameter tuning process highlights the effectiveness of the proposed model in accurately classifying disease instances.

4.4. Result analysis

In this section, we present the results of the experiments conducted. The overall performance of the proposed model was evaluated using the Liver diseases datasets. Additionally, we presented the ablation study of our experimental results for the proposed method.

4.4.1. Hyperparameter optimization

We investigated the impact of hyperparameter selection on the performance of AFLID-Liver for the binary classification of liver disease. The result is presented in Table 4. A grid search was conducted to evaluate configurations across varying dropout rates (0.1, 0.2, 0.3, 0.4, 0.5) and learning rates (0.001, 0.0003, 0.0005, 0.0008). Performance was assessed using key metrics including AUCROC, MCC, AUPRC, Accuracy, True Positive Rate (TPR), False Positive Rate (FPR), and True Negative Rate (TNR).

Across the range of tested hyperparameter configurations, the model demonstrated generally high performance, with metrics varying between: AUCROC (0.914–1.0), MCC (0.3706–1.0), AUPRC (0.7787–1.0),

Table 4

Hyperparameter optimization result of AFLID-Liver with robust scaler technique.

Dropout	Learning Rate	AUC-ROC	MCC	AUPRC	Accuracy	TPR	FPR	TNR	FNR	Training Time (s)	Inference Time (s)	Precision	Recall	F1-score
0.1	0.001	1.000	0.995	1.000	0.998	0.999	0.003	0.998	0.001	411.784	0.830	0.998	0.998	0.998
0.1	0.0008	0.999	0.996	0.991	0.998	0.999	0.002	0.998	0.001	549.071	1.030	0.998	0.998	0.998
0.1	0.0005	1.000	0.998	1.000	0.999	0.999	0.001	0.999	0.001	945.930	1.052	0.999	0.999	0.999
0.1	0.0003	1.000	0.996	1.000	0.999	0.995	0	1.000	0.005	1007.390	0.956	0.999	0.999	0.999
0.2	0.001	1.000	0.998	1.000	0.999	0.999	0.001	0.999	0.001	469.917	0.777	0.999	0.999	0.999
0.2	0.0008	0.999	0.998	0.999	0.999	0.997	0	1.000	0.003	541.423	1.006	0.999	0.999	0.999
0.2	0.0005	1.000	0.995	0.999	0.998	0.996	0.001	0.999	0.004	725.810	1.002	0.998	0.998	0.998
0.2	0.0003	1.000	0.988	1.000	0.995	0.990	0.003	0.997	0.010	951.854	0.771	0.995	0.995	0.995
0.3	0.001	1.000	0.992	0.999	0.997	0.995	0.003	0.998	0.005	419.110	0.842	0.997	0.997	0.997
0.3	0.0008	0.999	0.995	0.997	0.998	0.995	0.001	0.999	0.005	597.676	0.926	0.998	0.998	0.998
0.3	0.0005	1.000	1.000	1.000	1.000	1.000	0	1.000	0	953.503	0.846	1.000	1.000	1.000
0.3	0.0003	0.988	0.750	0.970	0.898	0.656	0.004	0.996	0.344	945.712	0.858	0.909	0.898	0.891
0.4	0.001	1.000	0.996	1.000	0.999	0.998	0.001	0.999	0.002	531.399	0.855	0.999	0.999	0.999
0.4	0.0008	0.999	0.993	0.997	0.997	0.995	0.002	0.998	0.005	576.508	0.759	0.997	0.997	0.997
0.4	0.0005	1.000	0.984	0.999	0.994	0.983	0.003	0.998	0.017	942.180	1.717	0.994	0.994	0.994
0.4	0.0003	0.988	0.825	0.972	0.929	0.769	0.007	0.993	0.231	938.829	1.007	0.933	0.929	0.926
0.5	0.001	0.999	0.991	0.998	0.996	0.996	0.004	0.996	0.004	572.058	1.016	0.996	0.996	0.996
0.5	0.0008	0.999	0.994	0.997	0.997	0.995	0.002	0.998	0.005	776.516	0.782	0.997	0.997	0.997
0.5	0.0005	0.999	0.995	0.998	0.998	0.995	0.001	0.999	0.005	935.439	0.919	0.998	0.998	0.998
0.5	0.0003	0.914	0.371	0.779	0.767	0.199	0.004	0.996	0.801	774.197	1.009	0.811	0.767	0.707

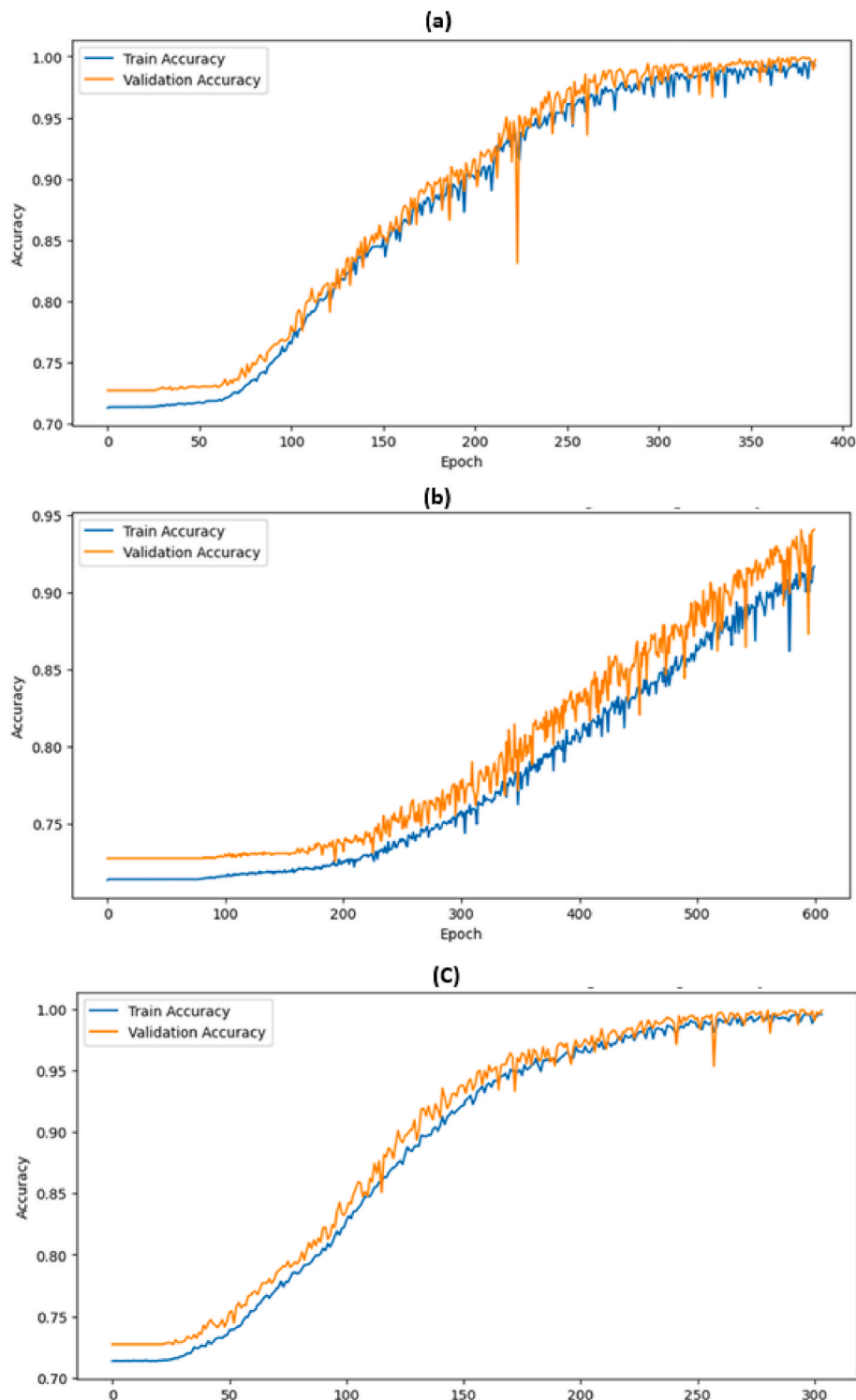


Fig. 7. Training and Validation curve of LSTM with Attention and Focal Loss using three different scaler: (a) Standard, (b) Min-Max (c) Robust.

and Accuracy (0.7671–1.0). TPR (recall) ranged from 0.1988 to 1.0, FPR from 0.0 to 0.0067, and TNR from 0.9933 to 1.0. The hyperparameter optimization successfully identified an optimal configuration at a dropout rate of 0.3 and a learning rate of 0.0005. This specific setting achieved perfect scores across all evaluated metrics: AUCROC=1.0, MCC=1.0, AUPRC=1.0, Accuracy=1.0, TPR=1.0, FPR=0.0, and TNR=1.0.

Analysis of the results revealed distinct trends associated with the interplay between dropout and learning rate. At lower dropout rates

(0.1–0.2), performance remained high, particularly with learning rates of 0.0005 and 0.001, often achieving near-perfect classification metrics (for instance, dropout 0.1, learning rate 0.0005 yielded AUCROC=1.0, MCC=0.9978, TPR=0.999, FPR=0.0008). However, higher dropout rates (0.4–0.5) showed increased sensitivity to lower learning rates. For instance, the combination of dropout 0.5 and learning rate 0.0003 resulted in a drastic performance decline, with AUCROC dropping to 0.914, MCC to 0.3706, and critically, TPR plummeting to 0.1988 while FPR remained relatively low (0.0042). This significant reduction in TPR

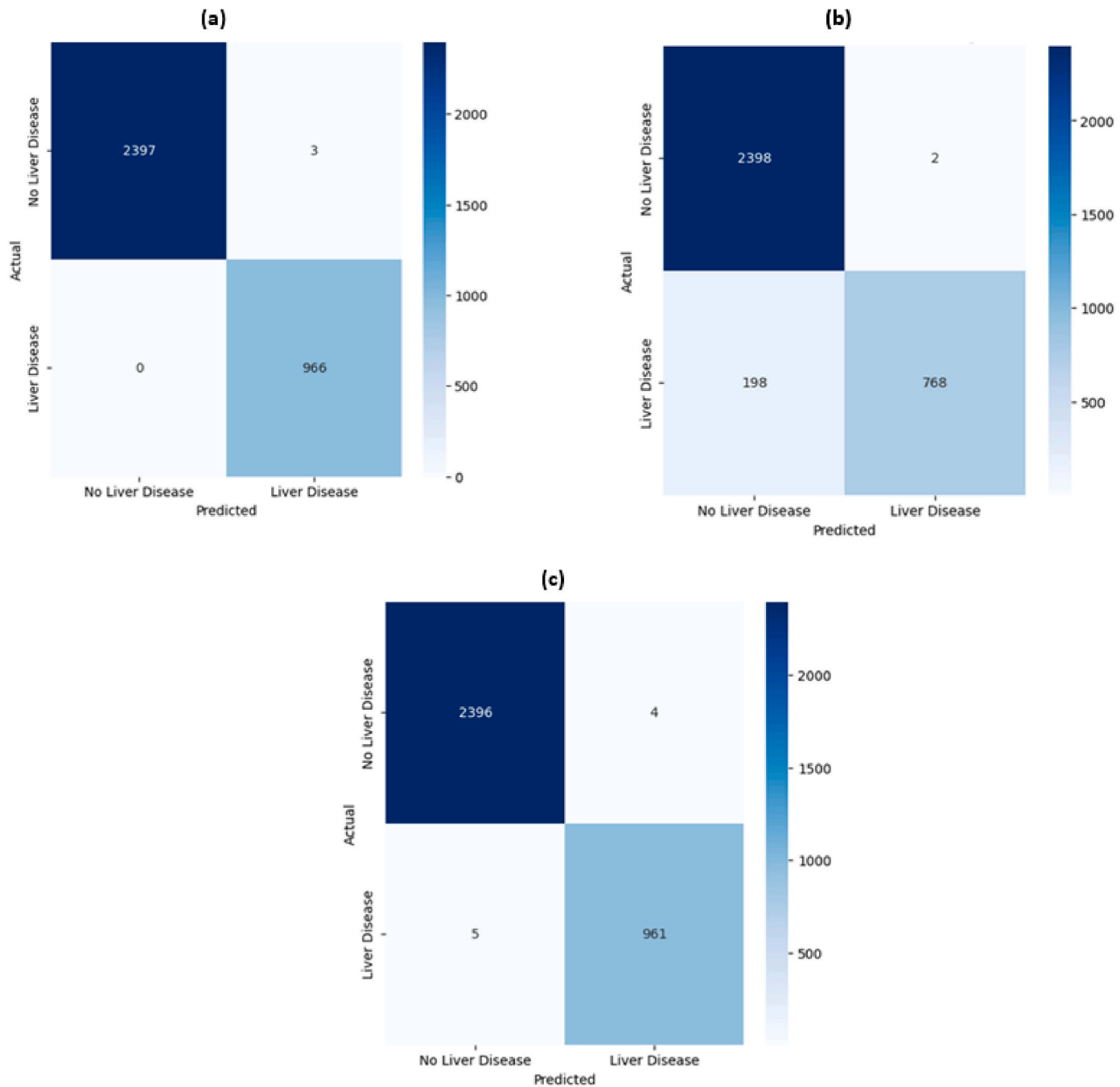


Fig. 8. Confusion Matrix of LSTM with Attention and Focal Loss using three different scalers: (a) Standard, (b) Min-Max (c) Robust.

indicates a high false negative rate, rendering such configurations unsuitable for clinical application where missing cases of liver disease is highly undesirable.

The optimal configuration, identified as dropout 0.3 and learning rate 0.0005, yielded perfect classification metrics on the evaluation set, demonstrating the potential to significantly surpass baseline performance by effectively balancing regularization and learning dynamics.

The sensitivity analysis highlighted that while moderate learning rates of 0.0005 is robust across varying dropout levels, a low learning rate combined with high dropout can lead to severe underperformance, particularly impacting the model's sensitivity (TPR). These findings underscore the importance of careful tuning to achieve optimal sensitivity and specificity, which are paramount for reliable medical diagnostic analysis.

Table 5
Distribution of result for scaling method.

Scaling Method	AUC-ROC	MCC	AUPRC	Accuracy	Precision	Recall	F1-Score	Training Time	Inference Time
standard	1.000	0.998	1.000	0.999	0.997	1.000	0.998	1674.373	2.032521
minmax	0.994	0.855	0.983	0.941	0.997	0.795	0.885	2544.885	1.456324
robust	1.000	0.993	0.999	0.997	0.996	0.995	0.995	1356.29	1.749238

4.4.2. Performance evaluation of the proposed model across various scaling methods

In selecting the best scalar method for the proposed model, this study also evaluated the impact of Standard, Minmax, and Robust feature scaling methods on the performance of AFLID-Liver Model. As depicted in Figs. 7 and 8, Analysis of training and validation accuracy curves across epochs revealed consistent convergence patterns and a clear indication of overfitting across all scaling methods. While training accuracy for all methods rapidly approached 1.0 (stabilizing between epochs 150–200, though with some fluctuations), validation accuracy consistently plateaued at approximately 0.75 after reaching this level by epoch 100. This persistent divergence between training and validation performance, representing a substantial accuracy gap, demonstrated that the model effectively memorized the training data but failed to generalize comparably to unseen validation data, irrespective of the scaling applied.

Evaluation on the test set using confusion matrices and performance metrics highlighted significant differences in how each scaling method affected the model's ability to classify liver disease. Standard Scaling resulted in minimal misclassifications (3 FP, 0 FN, 966 TP, 2397 TN), achieving high metrics including AUC ROC = 0.9999, Recall = 1.0000, and F1-score = 0.9985. Robust Scaling also performed strongly (4 FP, 5 FN, 961 TP, 2396 TN), yielding comparable metrics like AUC ROC = 0.9996, Recall = 0.9948, and F1-score = 0.9953, and notably had the shortest training time (1356.29 s). In stark contrast, Minmax Scaling led to substantially poorer performance, particularly concerning the positive class, with a high False Negative count (198 FN, 768 TP, 2 FP, 2398 TN) and consequently lower Recall (0.7950), F1-score (0.8848), and MCC (0.8553), alongside the longest training time (2544.88 s).

The results indicate that Standard and Robust Scaling enable the AFLID-Liver model to achieve high performance and strong discrimination on the test set, effectively minimizing false negatives which is crucial in a medical context. The use of focal loss appears effective in balancing predictions with Standard and Robust Scaling, where false negatives are low. However, Minmax Scaling significantly impairs performance, likely due to its sensitivity to outliers or data distribution effects that negatively interact with the model and focal loss, leading to a high false negative rate and reduced overall classification capability. Detailed summary of the result is presented in Table 5.

4.4.3. Overall performance of the proposed study

The performance of the models, AFLID-Liver (AM + LSTM + FL) and GRU, was evaluated in the context of liver disease classification using key metrics such as accuracy, precision, recall, F-score, False Alarm Rate (FAR), False Negative Rate (FNR), and Inference Time as shown in Table 6. These metrics provide insights into the models' ability to accurately detect and classify liver disease cases, which is critical for timely diagnosis and treatment. The AFLID-Liver model demonstrated exceptional performance, achieving an accuracy of 99.9 %, precision of 99.9 %, recall of 99.9 %, and an F-score of 99.9 %. Additionally, its FAR was 0.00, and its FNR was 0.001, indicating a near-perfect ability to classify liver disease cases while minimizing errors, as seen in Fig. 9. This suggests that the model is highly effective in identifying true cases of liver disease, while avoiding misclassification of healthy individual livers and missing actual cases. In comparison, the GRU model also performed well but was slightly less effective. It achieved an accuracy of 99.7 %, precision of 98.9 %, recall of 98.2 %, and an F-score of 97.9 %. However, its FAR of 0.13 and FNR of 0.11 indicate a higher likelihood of false alarms and missed detections compared to the AFLID-Liver model.

While these results are still commendable, they suggest that the GRU model may not be as reliable in critical liver disease classification scenarios, particular in an imbalanced scenario as illustrated by its reduced f-score of 97.9 % when compared with AFLID-Liver.

The superior performance of the AFLID-Liver model can be attributed to its advanced multi-model architecture. The Attention Mechanism (AM) enhanced the performance by identifying and prioritizing critical features associated with liver disease. The LSTM network effectively captures sequential dependencies and patterns in the data, while the Focal Loss (FL) addressed class imbalance, ensuring that minority classes, such as rare liver disease cases, are not overlooked during training. The GRU model, while simpler in design, still provides strong results. Its performance highlights its ability to capture temporal patterns and relationships in the data, but it lacks the additional enhancements provided by AM and FL, which likely accounts for its relatively lower performance.

The results however highlighted the AFLID-Liver model as the more robust and reliable choice for liver disease classification. Its ability to achieve near-perfect accuracy and minimize errors ensures that it can support timely and accurate diagnoses, ultimately improving patient outcomes. The GRU model, while a viable alternative, may be better suited for scenarios where computational simplicity is prioritized over peak performance as depicted in Fig. 9. The overall performance of the proposed model is presented in Table 6.

The training and validation accuracy trends for the AFLID-Liver model and the GRU model as illustrated in Figs. 10 and 11 demonstrate strong performance in the liver disease classification task, with both models achieving high accuracy levels. The GRU model exhibits a rapid improvement during the initial epochs, stabilizing at an accuracy of approximately 97.7 % after 50 epochs. Despite this, the model occasionally experiences sharp dips in validation accuracy, although it consistently recovers to maintain high performance. These fluctuations may be attributed to the model's complexity which can sometimes lead to overfitting. In comparison, the AFLID-Liver model achieves impressive accuracy, stabilizing around 99.9 % for both training and validation after 200 epochs. Unlike the GRU model, the AFLID-Liver model shows a smoother accuracy trend with fewer fluctuations in validation accuracy, suggesting better generalization to unseen data. This stability may stem from the model's robust architecture, which is likely less prone to overfitting while still capturing temporal dependencies effectively. While the AFLID-Liver model slightly outperforms the GRU model in terms of accuracy, the GRU model demonstrates lower inference time complexity. Both models are highly effective, but the choice between them may depend on whether achieving the highest possible accuracy, ensuring robust generalization, or where computational simplicity is prioritized for the classification task.

4.4.4. Ablation study

We carried out the ablation study and the experimental analysis obtained are given in Table 7. The evaluation of the different models as evaluated on the Liver diseases data revealed important insights into their performance. The ablation study conducted revealed significant improvements in both F-Score and False Positive Rate (FPR) for AFLID-Liver, having integrated LSTM with Attention mechanism and Focal Loss. The F-Score for AFLID-Liver was 99.9 %, showing a remarkable enhancement over the other models. Additionally, the False Positive Rate (FPR) for AFLID-Liver was minimized to 0.001 %, demonstrating a drastic reduction compared to the AM + LSTM (0.083 %) and LSTM + FL (0.032 %) models. These results indicate that the integration of attention

Table 6
Overall Performance of the proposed model.

Model	Accuracy	Precision	Recall	F-Score	FAR	FNR	Inference Time (s)
AFLID-Liver	99.9	99.9	100	99.9	0.00	0.001	1.43
GRU	99.7	98.9	98.2	97.9	0.13	0.11	1.08

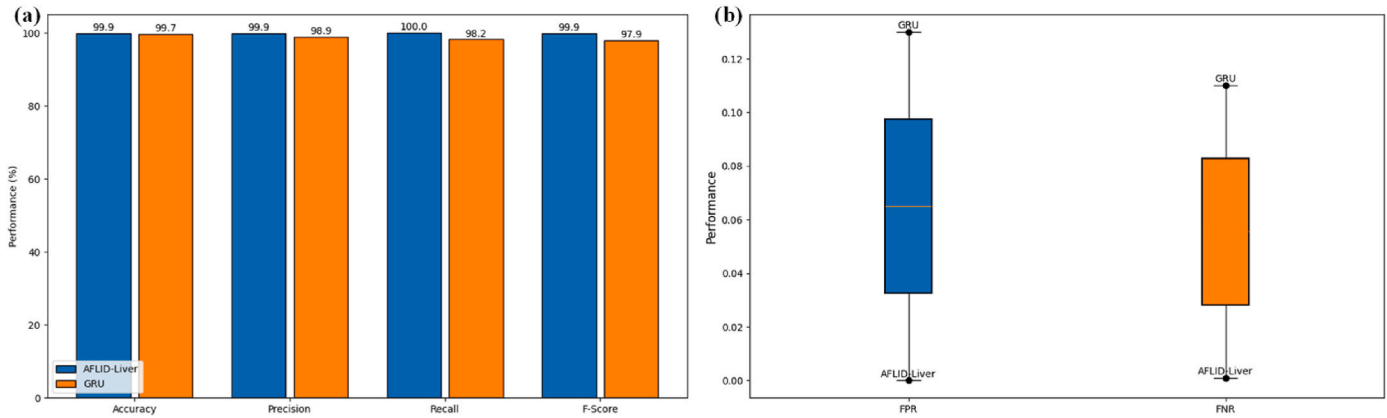


Fig. 9. Overall Performance of the proposed model.

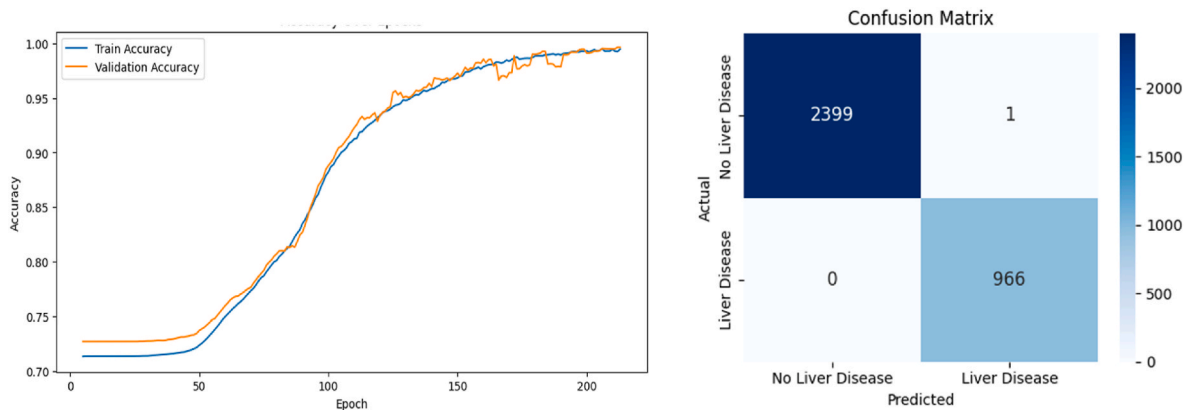


Fig. 10. AFLID-liver performance evaluation.

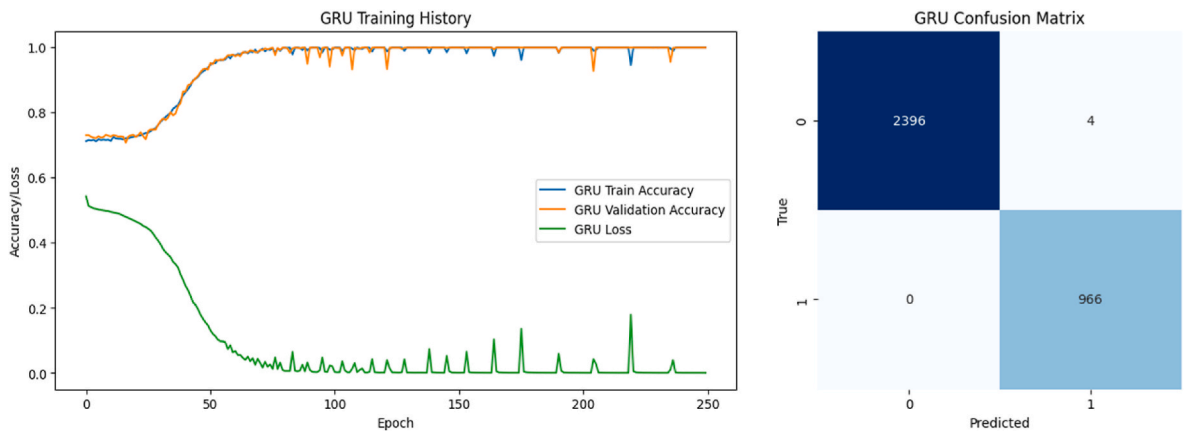


Fig. 11. GRU performance evaluation.

Table 7
Overall Performance of the proposed model and Ablation Study.

Model	ACC	PRC	RCL	F-Score	FPR	FNR	AUC	MCC	AUPRC	Training Time (s)	Memory Used (MB)	Inference Time (s)
LSTM	95.7	90.6	96.5	92.4	0.14	0.13	94.8	93.6	94.8	602.23	44.83	0.85
AM + LSTM	99.9	99.7	99.9	99.8	0.083	0.04	98.3	99.1	98.1	769.06	49.21	1.00
LSTM + FL	99.9	99.7	99.9	99.8	0.032	0.02	99.5	99.6	99.4	774.09	55.35	1.38
AM + LSTM + FL (AFLID-Liver)	99.9*	99.9*	99.9*	99.9*	0.001*	0.00*	99.9*	99.9*	99.9*	1175.37	62.66	1.43

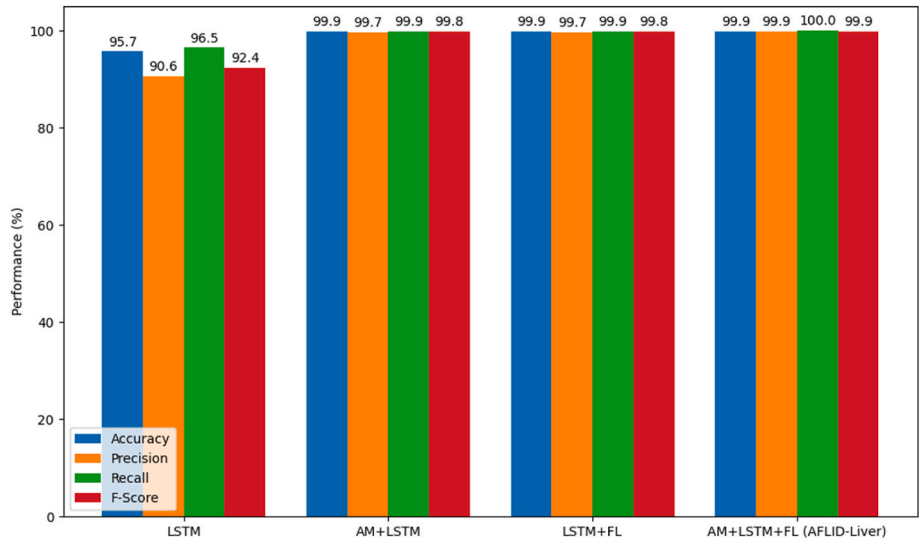


Fig. 12. Performance comparison of the ablation study.

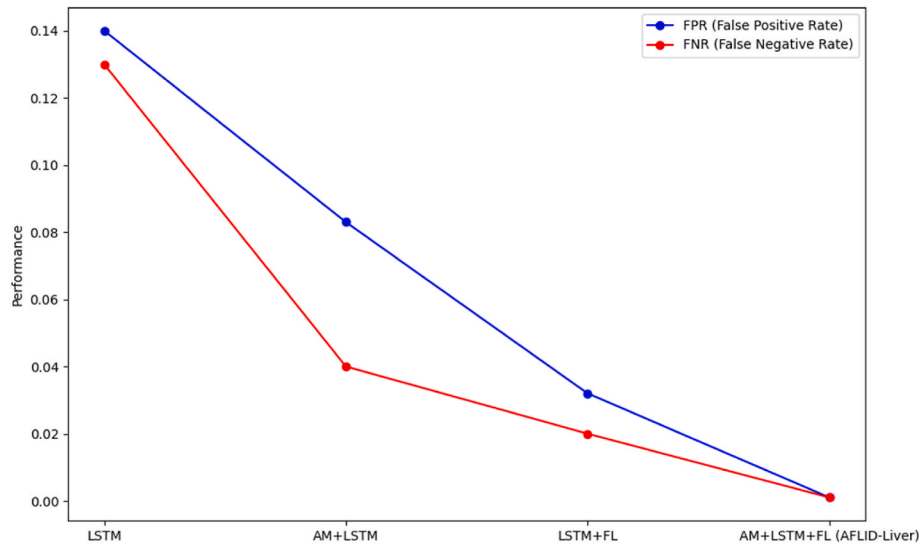


Fig. 13. False positive and false negative rate of the ablation study.

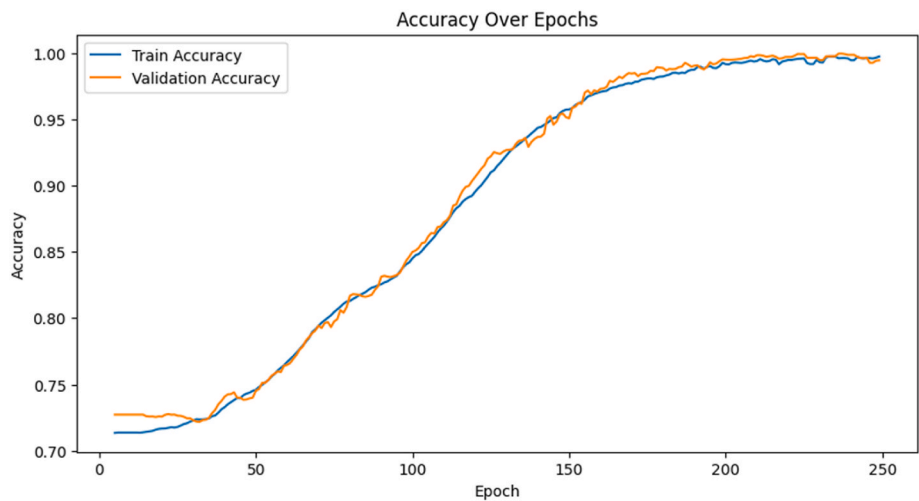


Fig. 14. Training and validation accuracy (LSTM).

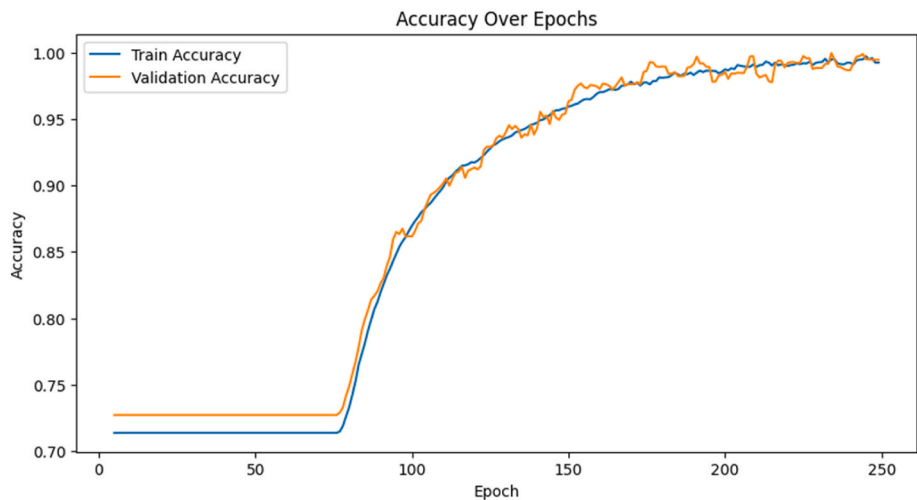


Fig. 15. Training and validation accuracy (LSTM with attention mechanism).

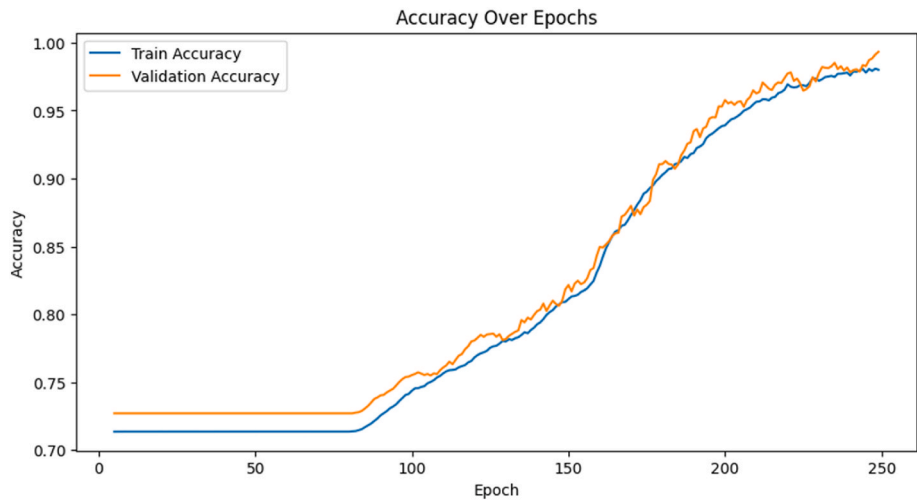


Fig. 16. Training and validation accuracy (LSTM with focal loss).

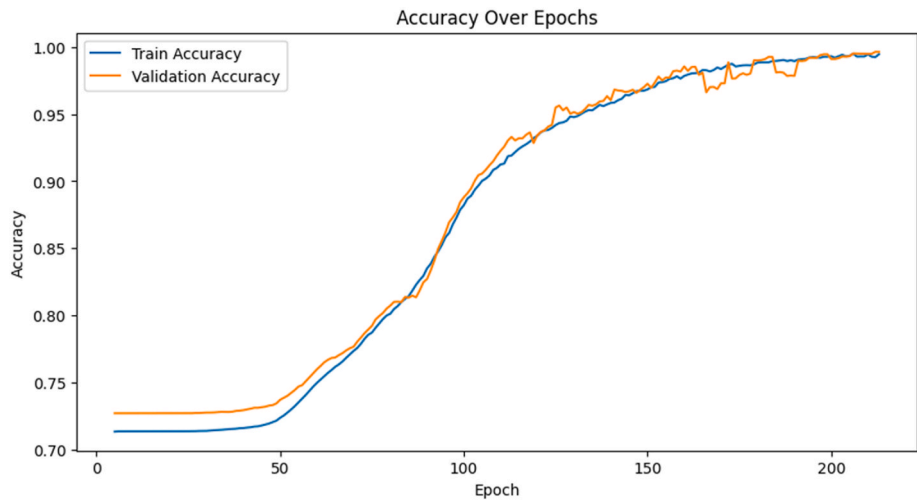


Fig. 17. Training and validation accuracy (LSTM with attention and focal loss).

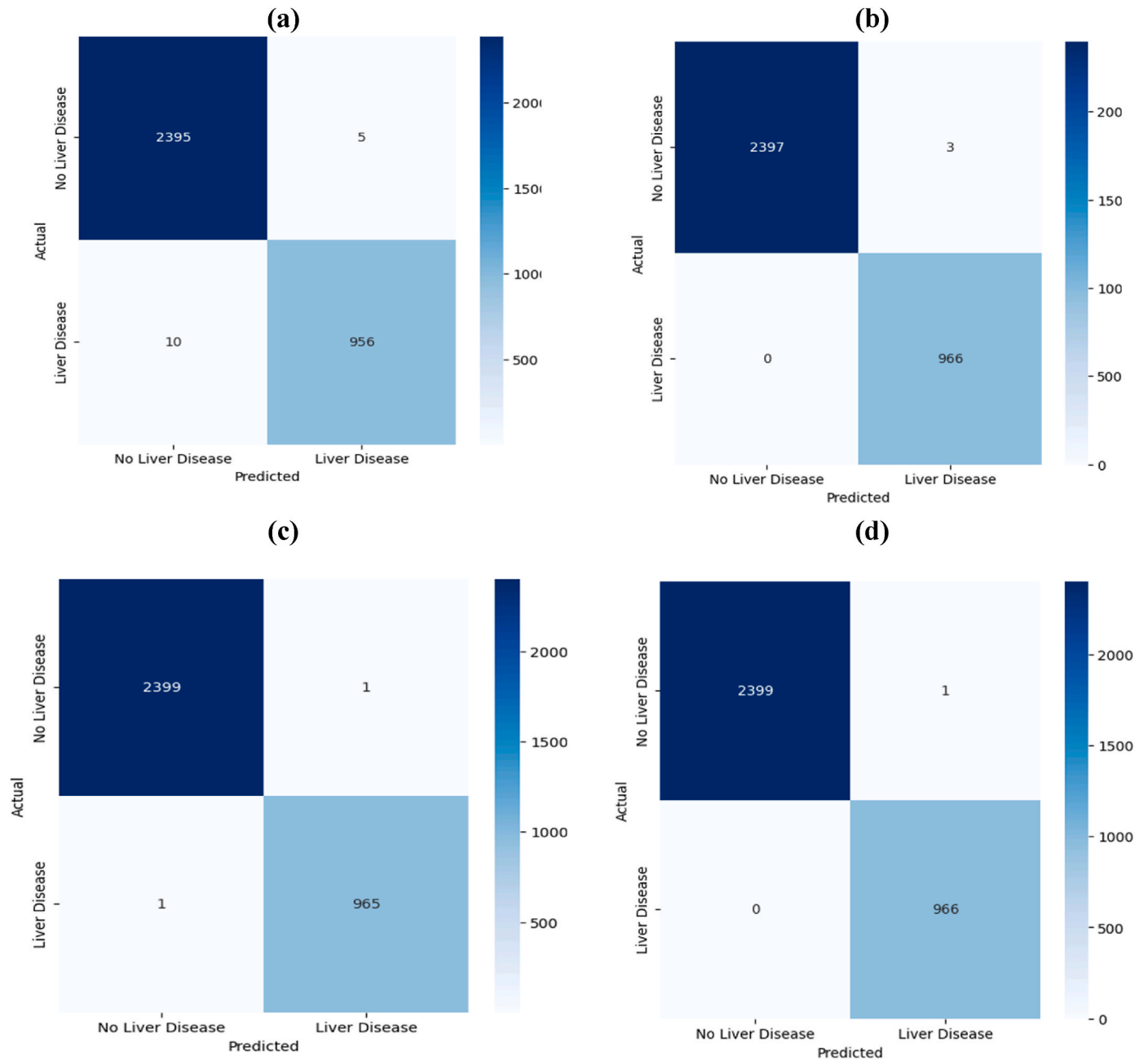


Fig. 18. Confusion matrix of (a) LSTM only, (b) LSTM + Attention, (c) LSTM + Focal loss, (d) LSTM + Attention + Focal loss.

Table 8

Cross-validation performance for AFLID-Liver.

Fold	Loss	Accuracy	Precision	Recall (TPR)	F1	AUC-ROC	MCC	AUPRC
1	0.0011	0.9994	1	0.9992	0.9996	0.9996	0.9985	0.9999
2	0.0003	0.9994	1	0.9991	0.9996	1	0.9985	1
3	0.0026	0.9991	0.9991	0.9996	0.9994	0.9987	0.9978	0.9994
4	0.0004	0.9997	1	0.9996	0.9998	1	0.9992	1
5	0.0016	0.9997	0.9996	1	0.9998	0.9991	0.9992	0.9992
Average	0.0012 (± 0.0009)	0.9995 (± 0.0002)	0.9997 (± 0.0003)	0.9995 (± 0.0003)	0.9996 (± 0.0002)	0.9995 (± 0.0005)	0.9987 (± 0.0005)	0.9997 (± 0.0003)

mechanism and focal loss in the AFLID-Liver model not only boosts classification accuracy but also significantly reduces the occurrence of false positives, enhancing its reliability for liver disease detection, although with high computational complexity compared with other baseline model. The comparison is captured and summarized in [Table 7](#),

[Figs. 12 and 13](#).

A comparative analysis of the training and validation accuracy for various model configurations in the ablation study was conducted to assess the effectiveness of the proposed AFLID-Liver model. The baseline LSTM model demonstrated relatively modest accuracy, with a slower

convergence rate and a visible gap between training and validation accuracy, indicating suboptimal generalization and potential overfitting as depicted in Fig. 14. Enhancing the model with an attention mechanism as illustrated in Fig. 15 improved convergence speed and led to better alignment between training and validation curves, highlighting the model's improved capacity to capture salient temporal features in the input sequences. Fig. 16 shows that the integration of focal loss alone into the LSTM architecture showed improved robustness to class imbalance, with enhanced validation accuracy and more stable training dynamics. However, the best performance was achieved with the AFLID-Liver model, which combines both attention and focal loss as depicted in Fig. 17. This model not only converged significantly faster than the others but also maintained consistently high accuracy across both training and validation sets. The minimal discrepancy between these curves reflects a strong generalization capability, while the integration of focal loss helped to mitigate the adverse effects of class imbalance. Overall, the AFLID-Liver model demonstrated superior predictive performance, learning stability, and convergence behavior, underscoring its suitability for complex, imbalanced classification tasks in liver disease analysis. Fig. 18 shows the confusion matrix for all these models.

4.4.5. Cross validation

The performance of the proposed model was cross validated using an external Liver diseases dataset sourced from <https://archive.ics.uci.edu/dataset/225/ilpd+indian+liver+patient+dataset>, the result is presented in Table 8. The proposed model performance was rigorously evaluated using 5-fold cross-validation. The dataset was partitioned into five subsets; in each iteration, the model was trained on four folds and validated on the remaining held-out fold. This process was repeated five times, with each fold serving as the validation set once, to provide a robust estimation of the model's generalization capability.

The cross-validation results demonstrate exceptional performance across a comprehensive suite of evaluation metrics. The mean values across the five folds were as follows: Accuracy 0.9995 (± 0.0003), Precision 0.9997 (± 0.0002), Recall 0.9995 (± 0.0003), F1-score 0.9996 (± 0.0002), AUC-ROC 0.9995 (± 0.0004), MCC 0.9987 (± 0.0008), and AUPRC 0.9997 (± 0.0002). The consistently high mean values, coupled with notably low standard deviations, indicate both high predictive performance and remarkable robustness of the model across different data partitions. Low training and validation loss values further confirm that the model effectively optimizes the Focal Loss objective.

The high AUPRC value (0.9997) strongly indicates the model's effectiveness in handling potential class imbalance within the dataset, a scenario where Focal Loss is particularly beneficial. The elevated AUC-ROC (0.9995) and MCC (0.9987) metrics highlight the model's near-perfect discriminative power, suggesting that the integrated Attention Mechanism successfully guided the LSTM to leverage the most relevant sequential features for accurate classification. These results collectively validate the synergistic benefits of the proposed model architecture and training methodology for this task.

4.4.6. Comparison with the state-of-the-art techniques

Table 9 compares the proposed AFLID-Liver model with existing approaches for liver disease classification, emphasizing the

methodological innovations and the underlying reasons for its superior performance. While previous studies applied a range of machine learning and deep learning techniques such as Naïve Bayes combined with Logistic Regression [42], Gradient Boosting [43,53], CNN + LSTM [44], Extra Tree Classifier [45], and Random Forest [46], they exhibit limitations in either capturing sequential patterns, handling class imbalance, or optimizing learning focus. In contrast, the AFLID-Liver model leverages a synergistic combination of Long Short-Term Memory (LSTM) networks and an Attention Mechanism, enabling the model to capture temporal dependencies in patient data while dynamically focusing on the most relevant features during training. This architecture ensures a better understanding of data patterns, especially in complex and noisy clinical datasets. Furthermore, unlike prior models that either applied no balancing technique or relied on oversampling approaches like SMOTE, AFLID-Liver integrates Focal Loss, a powerful loss function that inherently addresses class imbalance by emphasizing underrepresented instances which are hard to classify without introducing synthetic noise. This contributes to its robustness and enhances the model's sensitivity to minority class cases.

Performance-wise, AFLID-Liver significantly outperforms all existing methods across key metrics. It achieved 99.9 % accuracy, surpassing prior models whose accuracy ranged from 76.2 % [46] to 99.18 % [42]. Moreover, AFLID-Liver attains a perfect 99.9 % in precision, recall, and F-score, indicating exceptional balance between false positives and false negatives, which is something not consistently observed in previous models. Notably, it is one of the few models that reports False Negative Rate (FNR) and False Alarm Rate (FAR), recording extremely low values of 0.001 % and 0.00 % respectively, which further underscores its clinical reliability in avoiding missed diagnoses and false alarms. The methodological enhancements introduced in AFLID-Liver, namely the LSTM-Attention hybrid model and the application of Focal Loss, collectively contribute to its state-of-the-art performance. These elements address key challenges such as sequence learning, feature relevance, and class imbalance more effectively than traditional and existing hybrid models, affirming the AFLID-Liver model's robustness, accuracy, and practical applicability in liver disease detection.

5. Challenges, limitations, and recommendations

5.1. Resource overhead

The AFLID-Liver model, due to its combination of LSTM, Attention Mechanism, and Focal Loss, can be computationally intensive, especially during training. The model's complexity may require significant hardware resources and longer training times, making it less suitable for environments with limited computational power.

5.2. Scalability and efficiency trade-off

While the model performs well on the datasets used in the study, scaling it to larger datasets or real-time applications could pose challenges. The model's performance may degrade as the size and complexity of the input data increase. To enhance scalability, the model could be optimized for more efficient computation, possibly by

Table 9
Comparison of the Proposed Model with Existing work.

Models	Year	Methods	Balancing Technique	Accuracy	Precision	Recall	F-Score	FNR	FAR
[42]	2024	Naïve Bayes + LR	N/A	99.18	99.02	99.12	96	N/A	N/A
[43]	2024	Gradient Boosting	SMOTE	98.80	0.99	0.99	0.99	N/A	N/A
[44]	2024	CNN + LSTM	N/A	98.73	99	98	98	N/A	N/A
[45]	2023	Extra Tree Classifier	SMOTE	91.82	92.72	91.89	92.30	N/A	N/A
[46]	2022	Random Forest	N/A	76.2	71.7	84.3	77.5	N/A	N/A
[53]	2024	Gradient Boosting	SMOTE	98.80	98.50	98.50	98.50	N/A	N/A
AFLID-Liver	2025	LSTM + AM	FL	99.9*	99.9*	99.9*	99.9*	0.001*	0.00*

exploring model pruning or quantization techniques to reduce resource consumption without sacrificing performance.

5.3. Threat to validity

Few threats to the validity of the proposed Model have been outlined below.

5.3.1. External validity

The performance of the AFLID-Liver model may be limited to the specific datasets used in the study. While the model performs well on the dataset used in the study, its generalization ability on different, unseen datasets remains uncertain. Variations in data quality and feature distribution could impact its performance.

5.3.2. Internal validity

While the model combines LSTM, Attention Mechanism, and Focal Loss, it is difficult to isolate the individual contribution of each component to the overall performance. Confounding factors, such as dataset-specific biases or data preprocessing choices, may influence the results, making it hard to attribute improvements in performance solely to the proposed model's components.

5.3.3. Construct validity

The metrics used to evaluate the model (e.g., accuracy, precision, recall, F-score) may not fully capture the model's performance, particularly in real-world scenarios where other factors, such as interpretability or scalability, are crucial. The focus on these metrics might overlook other important aspects of model behavior that are essential for practical deployment.

6. Conclusions and future work

This research proposed and evaluated the AFLID-Liver model, which combines Attention Mechanism (AM), Long Short-Term Memory (LSTM), and Focal Loss (FL) for liver disease classification. The performance of the AFLID-Liver model was compared with a GRU-based model using key metrics, including accuracy, precision, recall, F-score, False Positive Rate (FPR), and False Negative Rate (FNR). The AFLID-Liver model demonstrated superior performance, achieving an accuracy, precision, recall, and F-score of 99.9 %, with an FAR of 0.00 and an FNR of 0.001. These results indicate its exceptional ability to accurately classify liver disease cases while minimizing errors. The AFLID-Liver model's success is attributed to its multi-model architecture, where AM enhanced critical feature prioritization, LSTM captures sequential dependencies, and FL addressed class imbalance by ensuring adequate representation of minority classes during training. In contrast, the GRU model, while achieving strong results with an accuracy of 99.7 % and an F-score of 97.9 %, exhibited higher FPR (0.13) and FNR (0.11), reflecting a higher likelihood of misclassification and missed detections. Despite its superior accuracy, the AFLID-Liver model showed occasional fluctuations in validation accuracy during training, possibly due to the model's complexity and sensitivity to overfitting. The GRU model, with its simpler architecture, demonstrated more consistent validation accuracy, suggesting stronger generalization to unseen data. These findings highlight the AFLID-Liver model as a robust and reliable choice for liver disease classification, especially in scenarios where achieving the highest possible accuracy is critical. The GRU model, on the other hand, offers a viable alternative for applications where computational simplicity and generalization are prioritized. Future research could explore potential optimization strategies such as model pruning and quantization techniques to enhance the stability of the AFLID-Liver model, enhance deployment efficiency, and optimize its computational efficiency, making it more suitable for real-time clinical applications. Also, lightweight transformer-based architectures can also be investigated as alternative design choices for future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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